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MAST20030: Differential Equations
Tutorial: Week 6

Question 1

Solve the following initial value problems using Laplace transforms:

(a) $y'(t) + 2y(t) = 4t, \qquad y(0) = 3.$

(b) $y''(t) - 2y'(t) + y(t) = \mathrm{e}^{2t}, \qquad y(0) = y'(0) = 0.$

Question 2

(a) Rewrite the following function using a Heaviside function and find its Laplace transform:

$$f(t) = \begin{cases} t\,\mathrm{e}^{-t}, & \text{if } 0 < t < 1, \\ \mathrm{e}^{-t}, & \text{if } t \geqslant 1. \end{cases}$$

(b) Use a Laplace transform to solve the following initial value problem:

$$y''(t) + y(t) = \begin{cases} 3 & \text{if } 0 \leqslant t < 4, \\ 2t - 5 & \text{if } t \geqslant 4, \end{cases} \qquad y(0) = 1, \quad y'(0) = 0.$$

Question 3

(a) Use Laplace transforms to solve the following initial value problem:

$$y''(t) + 3y'(t) + 2y(t) = 1 + \delta(t-1), \qquad y(0) = 1, \quad y'(0) = -1.$$

(b) Solve the following initial value problem in terms of the unknown function f:

$$4y''(t) + 4y'(t) + 17y(t) = f(t), \qquad y(0) = y'(0) = 0.$$

(c) Solve the following integro-differential equation for $y(t)$ where $y(0) = 0$:

$$y'(t) + 4y(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 1,$$

Hint: write the integral as a convolution of y with 1.